The empirical recurrence for OEIS A283632: recovering a conjecture that is not written down

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Abstract

OEIS A283632 records "Empirical recurrence of order 81", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 240$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283632 is "Number of nX6 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements.". It has offset 1 and begins

$$5, 516, 21963, 700534, 19517898, 490925804, 11651389723, 264077146748, \ \ldots$$

The entry states:

> Empirical recurrence of order 81 (see link above)

As of the "Last modified" line on the live entry (Mar 12 2017 14:31:17, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4004$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n\tau$, so they may be merged without changing any count; merging leaves $S' = 240$ of the 4004, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 240$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 480$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{81} c_i\, a(n-i),$$

c_1	45	c_{22}	−7409659093931709	c_{43}	−3650767763079464433339	c_{64}	−598614912189
c_2	−501	c_{23}	20005244154500880	c_{44}	1245814857420064975932	c_{65}	46715777434
c_3	−3102	c_{24}	70539180390768049	c_{45}	1208870846186880177320	c_{66}	5588493408789
c_4	68415	c_{25}	32965166292211578	c_{46}	−1194187870429653421428	c_{67}	−90386907811
c_5	−5721	c_{26}	−289385751755513460	c_{47}	−2574194066716026234888	c_{68}	−376754843193
c_6	−3244352	c_{27}	−703637666227008632	c_{48}	648279118254235037937	c_{69}	1363856721859
c_7	11522757	c_{28}	1723228207576946 13	c_{49}	4206008248600308815643	c_{70}	1698162324277
c_8	97708854	c_{29}	2955175639934759787	c_{50}	256614420544486611531	c_{71}	−95936613817
c_9	−537981983	c_{30}	3202467719021992004	c_{51}	−5384289996041837068232	c_{72}	−47992495605
c_{10}	−1607344752	c_{31}	−3708376503979418160	c_{52}	−1630043351989528726710	c_{73}	390811203031
c_{11}	15416051493	c_{32}	−15339452173510461447	c_{53}	552711098717650937925 9	c_{74}	734205502959
c_{12}	14082926139	c_{33}	−10734051984522793250	c_{54}	3786067974366871193199	c_{75}	−95147535624
c_{13}	−318808373565	c_{34}	31730523928630546020	c_{55}	−5020861747626538275714	c_{76}	−2867073800
c_{14}	−32543822589	c_{35}	61693071380742922716	c_{56}	−6398122674087129103839	c_{77}	130562119906
c_{15}	4875464271372	c_{36}	−18672921252669605321	c_{57}	4975030777136343926676	c_{78}	−4640796443
c_{16}	−1692928036029	c_{37}	−142476017510040233634	c_{58}	7588169690874534692676	c_{79}	−9475063065
c_{17}	−61860646594716	c_{38}	−104046969644294217375	c_{59}	−4735104573056559708090	c_{80}	3847851831
c_{18}	23106713671733	c_{39}	184702693423425505337	c_{60}	−6669544218676731994306	c_{81}	2936292331
c_{19}	674074344264876	c_{40}	414348302502122004753	c_{61}	2803135536485247446610		
c_{20}	206726323281120	c_{41}	−613849350168755916 99	c_{62}	5812091598576998351238		
c_{21}	−5235372364974042	c_{42}	−871892472279688974807	c_{63}	−492189900142762245502		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81+4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

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